

Rugby Sevens



Nick Poulos

Performance Demands of Rugby Sevens

In this chapter, we introduce the sport of Rugby Sevens, which is unique from Rugby Union (chapter 28), the 15-a-side game, and discuss the various performance factors and relative contribution that physical attributes contribute toward achieving performance and winning games, with particular emphasis on repeated performances over the typical tournament format.

Sport Description and Factors of Winning

Rugby Union Sevens (7s), also known as Rugby Sevens or Sevens Rugby, is a new Olympic sport that debuted at the 2016 Summer Olympic Games in Rio de Janeiro. Rugby 7s is a contact sport and considered a high-intensity variant of the traditional 15-a-side game. In 7s, teams are allowed seven players on the field at any time, with five players available for interchange. Elite international male teams compete across 10 tournaments between December and May each year as part of the Sevens World Series, in addition to major event tournaments (Olympic Games, Commonwealth Games, Rugby Sevens World Cup, each held every 4 yr). Games are typically played over a 2 d tournament, and occasionally over a 3 d

tournament. Matches consist of two 7 min halves, plus stoppage times, with a 2 min halftime interval on a full-dimension rugby field (maximum length of playing area = 100 m; maximum width of playing area = 70 m). Pool matches are held on day 1 to determine seedings for day 2, where teams compete for World Series points (20). The tournament usually consists of three group stage matches on the first day, each separated by ~2 to 4 h, and depending on results, up to 3 games on the second day of competition. The World Series is generally scheduled as sets of 2 tournaments in a row, in relative close proximity to one another, with sets separated by 4 to 6 wk, and with tournaments held in different countries, often on opposite sides of the world (20). Rugby 7s player positions are generally categorized into two distinct playing groups: forwards and backs. Recently, however, specific players often classified as hybrid players possess the ability to perform either position. Like Rugby Union (chapter 28), the object of the game is to score a try (5 points) by placing the ball across the opposition goal line and/or kicking the ball over a regulation Rugby Union goal post as a conversion following a try (2 points), penalty (3 points), or in general play (3 points). Basic rules mimic Rugby Union also, where players may use their hands or feet to move the ball, may pass the ball only in line or backward, and may not lose the ball forward while in motion.

Relative Contribution of Physical Performance

As with all team-based sports, especially those that are combative in nature or involve collisions, technical skills, tactical awareness, game knowledge, and experience are the main contributors to successful outcomes. Players with limited ability to pass the ball accurately, tackle, compete at the ruck contest or breakdown, or execute technical ability in a scrum or lineout will never compete at the elite level of Rugby 7s. In addition, circumstantial factors inherent with the game of Rugby 7s can prevent highly trained players the opportunity to exploit their physical qualities. Nevertheless, physical conditioning is extremely important in Rugby 7s to support the technical and tactical performance on the field, especially when under fatigue (figure 29.1). Physical conditioning, done properly, will support technical proficiency on the field. However, specific physical attributes are not always the difference between success and failure in Rugby 7s.

Given the ~45% greater relative game running distance/volume and ~135% greater high-velocity (>5 m/s) running demands than the 15-player rugby

game, international level Rugby 7s players are likely to have higher levels of endurance, lower levels of body fat, and higher lean mass profiles than 15-a-side Rugby Union players (15). While it is often proposed that additional body fat may act as a protective buffer in contact situations, excess fat will reduce a player's power-to-weight ratio, lower his ability to accelerate, and increase energy expenditure (12). Elite-level male Rugby 7s players have been reported to possess average skinfolds (sum of 7 sites) of 54.5 ± 11.6 mm. At the time of writing, our current squad skinfold average in late preseason is 40.5 ± 5.8 mm.

Elite-level Rugby 7s players are often selected based on their superior running speed and endurance qualities compared to 15-a-side players. Rugby 7s players have been reported to cover between 113 and 120 m/min, 45% more than Rugby Union players, and with a higher portion of that distance run at high velocity (>6 or 6.7 m/s) (15, 19, 24, 31). These running demands (speed and work:rest ratio) are higher than those previously recorded in Rugby Union (84 m/min with a work:rest ratio of 1.7:1) and rugby league (66 m/min with a work:rest ratio of 1.3:1) (11, 18, 29, 32). While caution is needed when



Figure 29.1 The position of the Rugby 7s player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging this varies slightly between positions (forwards, backs, hybrids). Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (40%), tactical awareness (20%), and physical capacities (40%) for Rugby 7s success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

comparing studies across different global positioning system (GPS) technologies (chapter 8), a 2012 study (15) of international players showed relative running intensities of 117 m/min with a work:rest ratio of ~1.8:1, while trained Spanish domestic Rugby 7s players have been reported as running at 107 m/min for backs and 98 m/min for forwards with a work:rest ratio of 1.7:1 (29). In comparing international against domestic matches in male 7s players, while total distance covered is similar, international matches are more intense, as shown by greater distances covered at higher velocities, along with more accelerations and decelerations (15). Higher distances at medium and sprinting speeds are also observed in backs relative to forwards in Rugby 7s (29). While Rugby 7s athletes are traditionally separated into forward and back positions, recently a hybrid profile athlete has emerged due to the specific skills and tactical elements required of the game. Indeed, players able to play a combination of both roles throughout a tournament may be important factors that enable Rugby 7s player success.

When comparing positions, game analysis suggests that Rugby 7s forwards spend 19% more time jogging, with backs spending 8% more time running (21, 23), while some have shown no clear delineation between positions (16, 24). Rugby 7s backs have been reported to cover more total distance and distance at medium (>14 km/h) and high speeds (>20 km/h) than forwards (30). However, forwards are involved in a higher number of scrums, tackles, and time spent at heart rate intensities >90% of maximum (HRmax).

In a World Rugby Sevens Series tournament, it is of note that team starters cover approximately 51% greater total distance and receive approximately 40% more contact efforts over the course of a tournament compared to backs in an 80 min 15-a-side Rugby Union match, despite similar playing durations (24, 28).

Basic anthropometric characteristics of Rugby 7s players have been reported as being substantially different than 15-a-side rugby players. International Rugby 7s backs tend to be ~2 cm shorter and ~6 kg lighter than international 15-a-side backs, with forwards ~1 cm shorter and ~13 kg lighter (16). Forwards are typically slightly larger and stronger than backs, possibly due to set-piece requirements, although the differences are not as significant as in 15-a-side rugby (19). International players have been reported as having greater upper-body strength and lower-body power and perform better during maximal

aerobic testing (effect size, 0.26 to 2.26) compared to provincial players, with mostly trivial-to-small differences between backs and forwards (25, 26). In addition, only trivial-to-small differences in physical match actions have been shown between forwards and backs, with forwards engaging in more contact situations and backs performing more passes, ball handles, and high-speed movements (28).

Rugby 7s players are often required to play a variety of positions during a match or tournament, with the exception of the scrum, performed by the half or hooker (28). Thus, Rugby 7s players as a team can generally be trained for similar match demands, with adjustments for individual player plans made based on individual physical capacities and requirements (28).

Targets of Physical Performance in Rugby 7s

The key targets of physical performance in Rugby 7s, illustrated relative to other sports in figure 29.1, include speed and metabolic conditioning (endurance).

Speed

Measures of speed have shown moderate and large relationships with attack-specific tasks within a match in international and provincial-level Rugby 7s players (26). Large relationships between 40 m sprint time ($r=0.70$) and number of line breaks and defenders beaten per match suggest that maximal speed is important to possess for elite Rugby 7s success. Similar relationships ($r=0.87$) have been shown with 10 m sprint times, suggesting that acceleration is also an important component of Rugby 7s players. Previous research (26) in international Rugby 7s players also showed that defensive and contact-specific match activities were related to sprint distance performances (5 m, 10 m, and 40 m), with all sprint distances showing moderate negative relationships with tackle score ($r=-0.32$ to 0.38) and missed tackles ($r=0.43$ and 0.47 , for 10 m and 40 m, respectively). In addition, sprint momentum at 10 m (body weight [kg] multiplied by sprint velocity [m/s]) also appeared to be a greater indicator of performance effectiveness than 10 m sprint time alone in contact situations, as evidenced by the stronger relationships shown between 10 m sprint momentum and effective defensive rucks ($r=0.36$ versus -0.09 , respectively) and dominant tackles

($r=0.31$ versus -0.14 , respectively) (26). In addition, a strong relationship ($r=-0.59$) was previously demonstrated between tackle score and repeat sprint (10×40 m) performance (26), suggesting the importance of being able to maintain speed under fatigue to consistently tackle effectively.

Metabolic Conditioning

Rugby 7s players at the elite level additionally require well-developed aerobic capacity and power to permit the repeating of these high-velocity actions over the course of a game and a tournament (30). Indeed, the ability to recover from high-intensity activities is especially important in Rugby 7s, as it is with many team sports (17). In addition, the ability to sustain very high-intensity activity over the duration of a match and tournament suggests that factors affecting fatigue and recovery are important considerations for performance (15). By way of illustration, international male Rugby 7s players have been reported to run an average of 1452 ± 243 m per match with an average of 252 ± 103 m accumulated at or above 5 m/s, with a mean peak velocity over 8 m/s (28). International Rugby 7s players have similar speed characteristics to 15-a-side backs across distances of 10 to 30 m. However Rugby 7s players possess superior aerobic endurance (24). As already outlined, Rugby 7s is characterized by relatively high running demands (speed, accelerations, changes of direction) with only short recovery periods between running bouts (15, 31). In addition, contacts encountered elicit greater internal loads (i.e., amplified HR responses) than just running alone (29). At the elite international level, match score line, opposition, and substitute timing have all been shown to influence the activity profile, with Rugby 7s players more likely to run hard against higher-ranked opponents and when the score line is close (20).

Key Weapons, Manipulations, and Surveillance Tools

Recall that weapons refer to the high-intensity interval training (HIIT) formats we can use to target the physiological responses of importance, while the surveillance tools are what we use to monitor the individual responses to those weapons, to tell us whether or not we achieved our target. In this section, the different HIIT weapons, their manipulations, along with ways to monitor their effectiveness (surveillance) in Rugby 7s are offered.

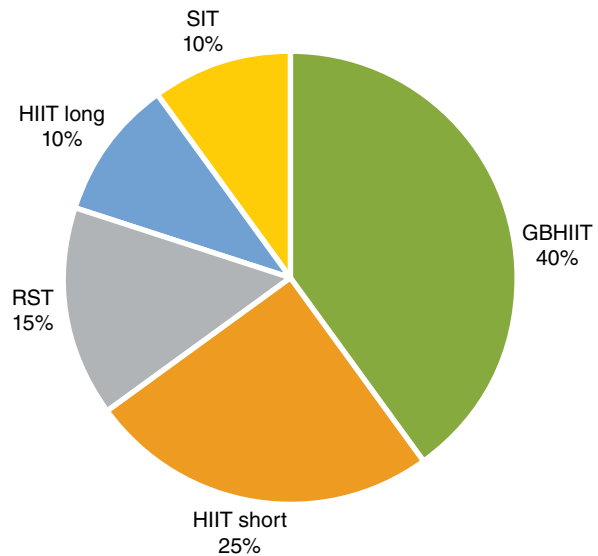


Figure 29.2 Percentage breakdown of the different HIIT formats (weapons) used throughout the annual season in elite international Rugby 7s.

HIIT Weapons

Recall that weapons refer to the HIIT formats we can use to target the physiological responses or target types. In the preparation of our national team, as previously outlined, we target all five HIIT types presented in figure 1.5 (5). For Rugby 7s, the large majority of these are opposed games, or game-based HIIT (GBHIIT), with rule variations or manipulation of attack and defense numbers, essentially variations of small-sided games (SSGs). This HIIT format is sometimes better known as manipulated numbers and/or rules (MNRs), and is completed on a full field or modified field size (40% both pre- and in season). The other weapons we use are short intervals (25%, some preseason but mostly in season), repeated-sprint training (RST) (15%, both pre- and in season), sprint interval training (SIT) (10%, mostly in season, some preseason, and for player rehabilitation or in-season conditioning top-up), and long intervals (10%, mostly preseason, some in season, some during specific stages of player rehabilitation, and some for in season top-up conditioning). The general breakdown of the HIIT formats used across a season for Rugby 7s is shown in figure 29.2.

Manipulations of Interval Training Variables

The running intensity and modality of each HIIT format we use is systematically manipulated to reach

desired acute metabolic and locomotor responses on specific days throughout the training week, which may vary between preseason and in-season periods. The unique nature of international World Series Rugby 7s tournaments (i.e., 3 to 4 wk turnaround between back-to-back tournaments) offers a unique programming puzzle we must aim to solve on a weekly basis. Adjustments in training are achieved through programming variations within and between the different days/weeks and cycles between tournaments. For example, if a high-intensity SSG or MNR rugby session is prescribed with speed injections (repeated all-out short sprints) targeting type 4, then a type 1 short-interval segment may be prescribed within or at the end of the training session, often in a straight line with slight 45° turns, as opposed to sharp change-of-direction repetitions to avoid overloading an already taxed neuromuscular system. This is particularly important as vital strength sessions are typically prescribed later in the training day. Alternatively, should a high-intensity SSG or MNR rugby session be prescribed, with no subsequent training session in the afternoon (e.g., a half day off) alongside recovery/rest days following, then a type 5 session may be prescribed within or upon completion of the session (e.g., 3 to 6 × all-out 30 s with 2 to 4 min recovery).

HIIT With Long Intervals (Outdoor)

We generally implement HIIT with long intervals during the off-season or very early preseason, usually on a rugby pitch (grass) or on a track, so as to minimize musculoskeletal strain (HIIT type 1 or 3 targets (6)). These typical HIIT sessions are generally performed over 2 to 3 min at ~85% to 93% of maximal aerobic speed (MAS), calculated from the average velocity attained over a 2 km time trial (1). This represents 490 to 580 m efforts completed over 2 min, depending on a player's fitness, with athletes running 3 to 6 repetitions interspersed by 2 min of passive recovery. For these, players are generally spread across 4 or 5 groups depending on their MAS. Sessions are generally prescribed 1 or 2 times per week, increasing to a frequency of 2 or 3 times prior to official preseason training. Such sessions also tend to be prescribed prior to a rest or recovery day, so that players can recover from an anticipated neuromuscular fatigue (see example in table 29.1). Importantly, this HIIT format is beneficial in that it evokes a large cardiopulmonary system response without reaching very high running speeds (<18 to 19 km/h). This is of primary importance in the context of the

weekly high-speed running load management, especially as a weekly high-speed running dose is prescribed and periodized in the final few weeks of the off-season in preparation for official preseason training. The prescription of such long-interval HIIT sessions lowers the locomotor and musculoskeletal overload, allowing other sessions to target this response with less risk of injury (14). We also use long-interval HIIT with players returning from rehabilitation or transitioning to structured training to ensure controlled straight-line running volumes and intensity. Here, our aim is to increase the chronic training load in a periodized or controlled manner to progressively enable return to play. Often, we may incorporate a ball (e.g., working in pairs or passing a specific set width) to allow players to practice their passing under the altered perception of load, as opposed to just straight-line long-interval running (see figure 29.3).

HIIT With Short Intervals

Our preferred short-interval HIIT weapons in Rugby 7s includes 10 s on/10 s off, 15 s/15 s, 20 s/20 s, and occasionally 10 s/20 s (e.g., figures 29.4 and 29.5), the latter format being low with respect to acute neuromuscular fatigue (figure 5.42 (5)). We often prescribe such HIIT short-interval sessions toward the end of a high-intensity rugby skills or tactical session. These repeated speed injections are used to add volume and intensity of locomotor load (i.e., high-speed running and mechanical work) so that neuromuscular load and anaerobic contribution can be carefully manipulated in consideration of future training sessions. Thus, type 1 short-interval HIIT is often used. In the latter stages of preseason, HIIT short intervals may include 20 s on/20 s off or 30 s on/30 s off (type 3), mostly straight line, with gradual introduction of change of direction or shuttles, again depending on the running and mechanical work that was part of the main rugby session.

Type 1, 2, 3, or 4 targets can all be hit with short intervals for Rugby 7s training. We use a number of these HIIT formats during the preseason within the training session, or at the end of the session, as stand-alone running. While we occasionally use short-interval HIIT formats during the preseason, these intervals are most often used in season for both team-based supplementary running or for players requiring specific individualized locomotor loads. A classic example of their use is in rehabilitating players or transitioning players from rehabilitation into the main high-intensity rugby drills or training sessions.

Table 29.1 $\dot{V}O_2$ and Strength/Speed Development Cycle
(General Phase, Earlier Preseason; Typically 3-4 Weeks Duration)

Microcycle		Monday	Tuesday	Wednesday
$\dot{V}O_2$ + strength/ speed develop- ment (generic)++	a.m.	Speed qualities	Strength: UB core lifts	Speed qualities
		Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets $\times$ 60 s to 2 min, 1 min rest depending on volume and periodiza- tion; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Contact conditioning	Rugby (contact): low-volume running load
		Conditioning: RST type 4 or 5; 3-6 $\times$ 30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodization of VHIR (team and individual); mostly straight line, hamstring load		Conditioning: repeat effort off-ground conditioning with repeat acceleration over 10-20 m (type 4)
		HIR short interval: type 2 (20 s/20 s) and type 3 (30 s/30 s) mostly straight line or gradual introduction of COD/ shuttle into next phase of preseason		
	p.m.	Rugby: craft session or low-intensity/ low running-volume skills	Rest	Speed qualities (acceleration): careful consideration to volume and type of type 4; repeat effort/ acceleration in the morning session and time between morning field session and after- noon session
		Strength: LB or WB core lifts		Strength: LB complexes with acceleration sprints (10-20 m indoor, 3-6 repetitions)

Note that two days are allowed after the lactic session Friday and that the long-bout HIIT is programmed before the day off. Type refers to the physiological target.

COD: change of direction; core lifts LB=squat variations, trap bar variations; core lifts WB=whole-body clean variations, snatch variations, deadlift variations, selected posterior chain exercises; core lifts UB=bench press variations, overhead pressing variations, vertical or horizontal pulling variations; HIR=high-intensity running (>5 m/s); VHIR=very high-intensity running (>6.7 m/s); contact conditioning=rugby-specific contact skills (e.g., wrestling variations); SSC=stretch shortening cycle.

Here, collective GBHIIT may not be either recommended or the players may not be ready to undertake such high-speed running or mechanical work. For rehabilitation players or transitioning players, we may also start with type 1 HIIT before progressing, depending on the type of injury, toward hitting type 2 targets (tailored toward either more high-speed running versus mechanical work), followed by type 3 targets, and finally, type 4.

In practice, we generally divide our players into 4 or 5 groups (19 km, 19.5 km, 20 km, 20.5 km, and >21 km for V_{IFT}) and request they run over group-based distances using cones on the rugby field. For example, for players with a V_{IFT} of 20 km/h, and for a 20 s/20 s HIIT run at 95% of V_{IFT} (relief interval: passive), the goal distance would be $(20/3.6) \times 0.95 \times 20 = 106$ m (20 is divided by 3.6 to convert the

speed from km/h to m/s, for convenience) (3). When we plan runs with changes of directions (CODs) to lower the volume of high-speed running and associated mechanical work, the time needed for COD must be considered when setting the goal run distance to ensure a related cardiorespiratory load compared to straight-line runs.

To make HIIT with short intervals more appealing to players and a bit more rugby specific in terms of movement patterns and locomotor loading, the ball is often integrated within the conditioning set in a variety of ways and often with a rugby-specific drill included within the set of the short-interval HIIT. For example, players run following position-specific running patterns for the required duration while reproducing position-specific technical sequences, including passes or defensive patterns or structures

Thursday	Friday	Saturday	Sunday
Rest	Speed qualities	Rugby: low speed clarity	Rest
	Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets × 60 s to 2 min, 1 min rest depending on volume and periodization; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Strength/cross training: individual specific needs	
	Conditioning: RST type 4 or 5; 3-6 × 30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodization of VHIR (team and individual); mostly straight line, hamstring load		
	HIRT short interval: type 2 (20 s/20 s) and type 3 (30 s/30 s) mostly straight line or gradual introduction of COD/shuttle into next phase of preseason		
	Rugby: craft session or low intensity/low running-volume skills	Rest	
	Strength: WB or LB power or complexes/plyometrics; fast SSC activity (late preseason and lower volume)		

(e.g., figures 29.4 and 29.5). Merging specific offensive or defensive movement patterns into short intervals is also an addition to our HIIT short intervals.

We also often insert an HIIT short-interval block between sets of rugby-specific defense attack drills. For example, 2 × 2 min periods of a 7 versus 7 attack defense drill may be followed by a 4 to 6 min HIIT short-interval segment of 15 s on/15 s off before another 2 × 2 min period of 7 versus 7 occurs. Often this is undertaken in straight-line format to avoid increased mechanical work or with a 45° turn to reduce neuromuscular load (type 1), unless sharper turns (90° to 180°) have been prescribed (increased neuromuscular requirements associated with deceleration and acceleration phases, increasing mechanical work, shifting to type 2 or 4 targets). Alternatively, an HIIT short-interval session can be given at the

end of the session for select individuals or the group (mostly type 1 for reasons previously stated) depending on periodized volumes (total volume, high-intensity or very high-intensity running volumes) or to ensure some players have received the appropriate dose after feedback using live GPS throughout the training session (see surveillance section below).

Repeated-Sprint Training

For our RST (type 4 target), we usually implement 1 or 2 sets of 3 to 6 × 30 to 40 m sprints, with 12 to 25 s of passive or active (jog or ~45% V_{IFT}) recovery (6) during the session, depending on load and periodization of very high-intensity running (VHIR), aiming to achieve >6.7 m/s (team and individual targets). Another variation of RST blocks is often

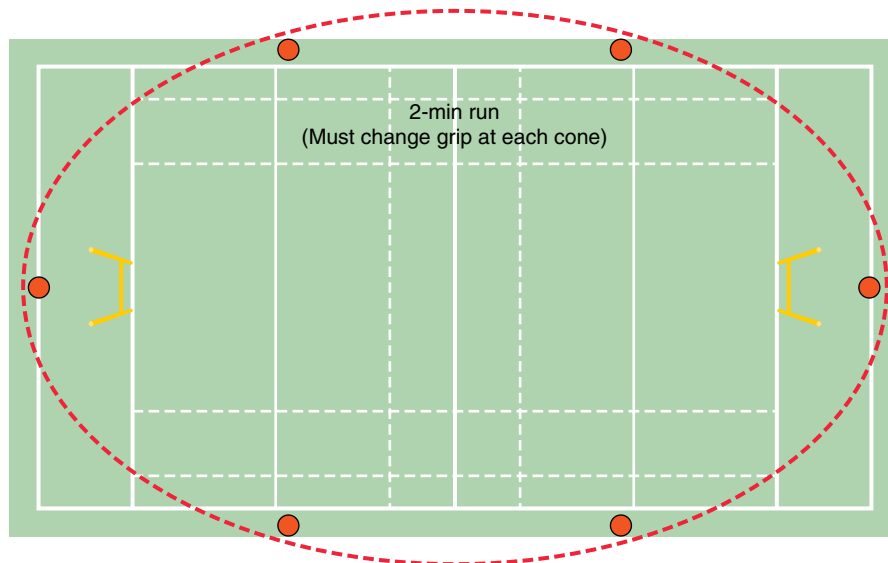


Figure 29.3 Example of a 2 min long-interval HIIT (type 3 target) with 2 min passive rest using the ball. For this session, we perform 2 min of straight-line running with the ball (individual distances based on 85% to 93% MAS from 2 km time trial). After passing each cone, grip is changed as follows: right arm (RA) with ball on chest, left arm (LA) with ball on chest, RA with ball on wrist, LA with ball on wrist, RA with ball in hand, LA with ball in hand. Following 2 min of passive rest, we repeat for 3 to 6 repetitions, depending on a number of variables.

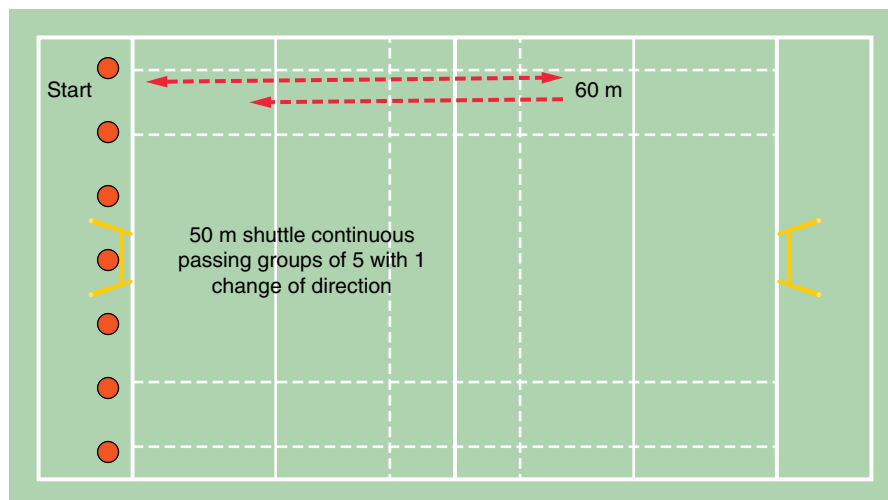


Figure 29.4 For this short-interval HIIT drill, we use a 20 s run over individual distances at 95% of the peak velocity achieved in the 30-15 IFT (see text, V_{IFT}). For a player with a V_{IFT} of 20 km/h, distance covered would be 1×60 m shuttle plus 42 m (102 m) as an example. It should be noted that this short-interval drill is placed within or at the end of a training session. V_{IFT} intensity may need to be adjusted (e.g., reduced to 92%-93%) due to accumulated fatigue in the training session. To begin the drill, players start to spread out across the field (approximately 5-8 m apart), use 20 s rest intervals, and use continuous passing for 50 m, including coach variations of 5 to 15 m passing widths with one shuttle run at ~92% V_{IFT} . This short-interval type 2 HIIT sequence is repeated for 6 to 12 min, depending on planned volumes, periodization, and placement within the training session. Sequence would be (i) Passing shuttle 50 m out and back; (ii) 20 s rest; (iii) 20 s run (1×60 m shuttle plus 42 m back using this example; (iv) 20 s rest; Repeat.

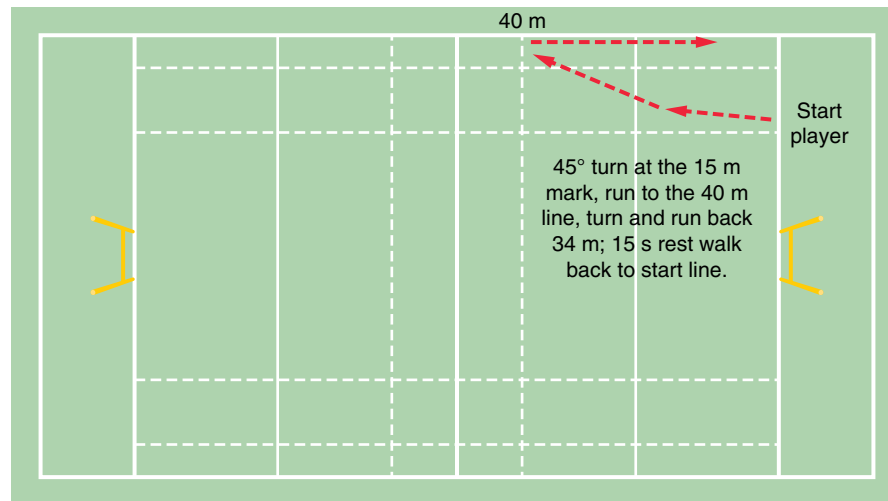


Figure 29.5 This short-interval sequence uses 15 s shuttle runs with 15 s passive (walk to target) recovery. During the 15 s shuttle, we simulate a press to shape the defense (running at $\sim 45^\circ$) to the 40 m mark (before turning and running back toward the start). We run these at 92% to 97% V_{IFT} (adjusting individual $\%V_{IFT}$ from 95% V_{IFT} for 45° angle and 180° shuttle). For someone running at 93% V_{IFT} (20 km/h), the player starts on the goal line, runs to the 15 m mark before turning $\sim 45^\circ$ and running until the 40 m marker, before shuttling and running back to about the 34 m mark, to walk or jog for 15 s to the start. The short-interval type 4 HIIT sequence is repeated for 5 to 8 min depending on planned volumes, periodization, and placement in the training session.

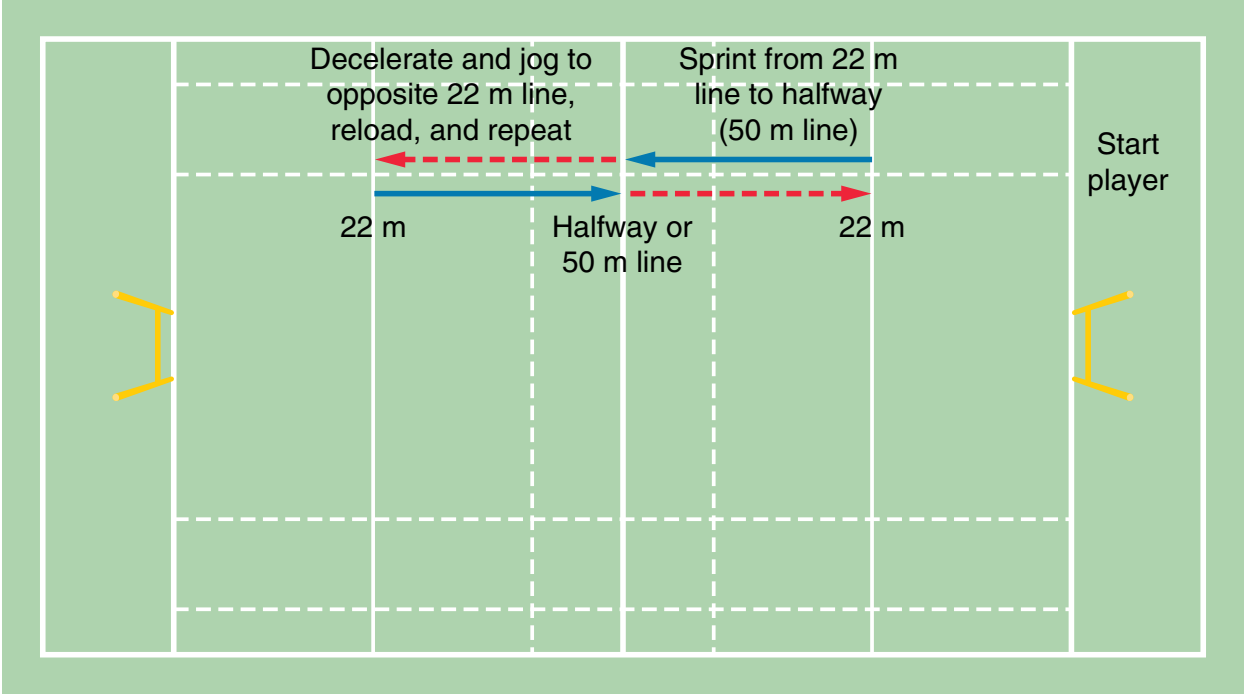
implemented as 2 sets of 5 s efforts/15 s active recovery over 4 min (figure 29.6) (5). We run these between the 22 m and halfway line, with a 15 s active recovery jog to the opposite 22 m line) (<https://www.youtube.com/watch?v=7aQb9M7ky7k>). These speed injections are mostly performed in the straight-line format, which come with the associated high hamstring load to be mindful of. We also implement an RST format of progressive rolling 30 m sprints or flys (10-20 m roll into 30 or 40 m fly) at progressive intensities early in the session following an adequate warm-up with the aim of reaching speeds >9 to 9.3 m/s, typical of speeds attained in a match but undertaken in a controlled manner in a nonfatigued state during the session. This allows an adequate dose of high-speed exposure (as an injury prevention tool) prior to commencing the main very high-intensity rugby drills throughout the forthcoming session.

As mentioned, we often intersperse these RST blocks between two training sequences as follows: warm-up; 5 min of technical work, speed technique, or speed efficiency; 5 to 10 min of another technical session; RST 1, which might include 20 m rolls progressing to 30 m fly sprints (>9 m/s), followed by walk recoveries (60-90 s) before 10 to 20 min of technical sequences; speed injection 1; 10 min of another technical sequence, followed by an HIIT

short interval block of 4 to 6 min (i.e., figure 29.5); finishing with another technical sequence. The programming of RST blocks should be carefully considered and executed with respect to forthcoming sequences or training sessions due to their substantial anaerobic and neuromuscular contributions (see tables 29.2 and 29.3 for more detail).

Sprint Interval Training

We typically use SIT in the form of 3 to 6 $\times$ all-out 20 to 30 s efforts with 2 to 4 min of passive recovery (type 5 target) (6, 8). As previously mentioned, SIT weapons are considered applicable when we have a clear afternoon schedule without a strength session planned and usually when a rest or recovery day follows. We may also insert the SIT weapon within a planned rugby-specific session. The SIT weapon is prescribed mostly in the final preparation period in our preseason once an adequate tolerance to high-speed running volume is attained. Given the intensity and duration of the format, it is difficult to intersperse rugby-specific work during these drills. Nevertheless, these can be added to the end of a session. Due to the high anaerobic contribution of SIT, strongly engaging the anaerobic glycolytic pathway alongside a large neuromuscular load and subsequent fatigue (22), its programming logic



	A grid	B grid	C grid	D grid	E grid
Distance	25 m	28 m (22-m line)	31 m	34 m	37 m
+ 1 (plus 1 stride more than the marker)	A + (26 m)	B + (29 m)	C + (32 m)	D + (35 m)	E + (38 m)
– 1 (minus 1 stride less than the marker)	A – (24 m)	B – (27 m)	C – (30 m)	D – (33 m)	E – (36 m)

Figure 29.6 As adapted from the work of John Pryor, SpeedPowerPlay (<https://www.youtube.com/watch?v=7aQb9M7ky7k>), this sequence of speed injections, which hits type 4 targets, involves 5 s sprints from the 22 m line to the halfway line, followed by 15 s active jogging to the opposite 22 m line. Players perform 4 to 6 repetitions depending on planned high-intensity running volumes, periodization, and placement within the training session. Note: faster or slower players can be grouped based on the four distance variations shown, which can be used to manipulate the training dose between sessions or even sets according to the situation. The field diagram illustrates a typical B grid.

applied during the sessions and during the week is similar to that described above for RST (see tables 29.2 and 29.3).

Game-Based High-Intensity Interval Training

We tend to plan and design GBHIIT in the form of opposed drills involving a varied number of players, with more or less numbers in attack or defense depending on the tactical purpose of the session or the target type of the drill. Most opposed sessions

involve variations of 7 versus 7 or 7 versus 6 (7 attackers versus 6 defenders), in which defenders are required to cover greater distances at consequently a high (>5 m/s) or very high running intensity (>6.7 m/s) in a full-field scenario. Other variations may include 10 versus 7 (10 attackers versus 7 defenders), in which players are required to defend a 100 m width within only 70 m of attacking territory.

While many practitioners in Rugby 7s use a variety of conditioning games or SSGs in their programming, we prefer to use mainly one common

drill, termed offside touch, in which physiological and locomotor loading can be adjusted using variations of field dimensions and player number rules (as above). Another specific example of a GBHIIT we use is based on an unstructured random game-based defensive situation in which 4 players must continuously defend against 7 or 8 attackers for 60 or 90 s. For this GBHIIT, defenders are required to plug random defensive holes of the attacking wave, immediately reset within a grid (see table 29.4 for variations in dimensions and associated impact on physical parameters) before another attacking wave is launched from another random side of a grid. The number of times the defenders are able to plug and prevent an attacking wave from scoring within the allocated time is counted as the competition element for the teams/sides.

Figure 29.7 provides a comparison between a typical SSG (offside touch) and tactical drill (7v7) in comparison to typical HIIT sessions and specific game demands. Such a comparison allows us to plan and periodize specific elements of our training over multiple weeks. As shown, the SSG and tactical drill provide lower accelerations and decelerations per minute compared to the tactical 7v7 drill and peak periods during a typical match.

Load Surveillance and Monitoring Tools

With the recent advent of GPS, inertial sensors, and semiautomatic video systems, the use of wearables and tracking technologies is common in Rugby 7s, during both training and matches (10, 24, 30). In our practice, we measure individual daily locomotor load tracking potential changes over time (i.e., acute/chronic ratio (14)). Despite their known limitations (chapter 8), we also tend to monitor HR during most training sessions that include a metabolic component, like HIIT.

Training Status Surveillance and Monitoring Tools

In alignment with the recommendations in chapter 9, the training status monitoring tools that we use in Rugby 7s generally involve simple, noninvasive, and inexpensive measurements, including overall well-being (27), jumps (neuromuscular status (5)), exercise HR (fitness (4) using a submaximal [12 km/h] HR test (9)), and GPS/accelerometer-derived data (including both locomotor loads and mechanical work (7)).

We have more recently used HR responses to HIIT sequences as general relevant markers of training adaptation (the lower, the better the player's fitness when comparing similar individual sequences) (9). We also use the training responses to assist us with decisions regarding an individual player's subsequent top-up HIIT training or cross-training, as required.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in Rugby 7s may be confined by various controllable and uncontrollable factors, before proper progression and periodization can be appropriately implemented. This section discusses some of the nuances of training prescription and HIIT structuring in Rugby 7s.

Controllable and Uncontrollable Factors

In our experience, the only truly uncontrollable factors within the planning and design of the Rugby 7s training program are the tournament locations and game schedule timing, which are controlled by World Rugby. In addition, gym-based strength and power training, in addition to high-quality technical sessions (e.g., handling drills, scrum and lineout technical and tactical sessions) together can constrain the timing, volume, and objective (HIIT type) of the HIIT sequences within a given week, as well as between tournaments. The chosen HIIT sessions within or at the end of a high-intensity Rugby 7s training session need careful consideration, with the aim of minimizing any neuromuscular strain post-HIIT and potential interference effects (chapter 6) should the subsequent strength and power session be performed on the same day, as can occur in our program. For example, some HIIT sequences require a large amount of mechanical work (i.e., types 2 and 4) on top of high-intensity rugby sessions, which may include the previously mentioned speed injections. Therefore, any undertaken HIIT training should be prescribed with the aim of minimizing subsequent neuromuscular strain (i.e., type 1), given a strength and power session and possibly a low-volume tactical running or technical session may be completed several hours later in the day. In addition, just as with the sport of elite soccer (chapter 30), the performance of a high-intensity rugby training session that incorporates HIIT can lower a player's attention to detail

Table 29.2 $\dot{V}O_2$ and Strength/Speed Development Cycle (Specific Phase 1, Late Preseason; Typically 4 Weeks During or Occasionally as a Heavier Preparation Week Between Tournaments)

Microcycle		Monday	Tuesday	Wednesday
$\dot{V}O_2$ + strength/speed development (specific)++	a.m.	Speed qualities	Strength: UB core lifts	Speed qualities
		Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets $\times$ 60 s to 2 min, 1 min rest depending on volume and periodization; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Contact conditioning	Rugby (contact): low-volume running load
		Conditioning: RST type 4 or 5; 3-6 $\times$ 30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodization of VHIR (team and individual); mostly straight line, hamstring load		Conditioning: repeat effort off-ground conditioning with repeat acceleration over 10, 20, or 30 m (type 4); consider lactate**
		HIR short interval: straight line (type 1) or COD (type 2) 10 s/10 s or 15 s/15 s/shuttle depending on focus for strength session in p.m. (avoid mechanical load); mostly straight line as neuromuscular already overloaded in high-intensity game simulated segments during training		
	p.m.	Rugby: craft session or low intensity/low running-volume skills	Rest	Speed qualities (acceleration): careful consideration to volume and type of type 4; repeat effort/acceleration in the morning session and time between morning field session and afternoon session
		Strength: LB or WB core lifts		Strength: LB complexes with acceleration sprints (10-20 m indoor, 3-6 repetitions)

Note that two days are allowed after the lactic session Friday and that the long-bout HIIT is programmed before the day off. Type refers to the physiological target.

COD: change of direction; core lifts LB=squat variations, trap bar variations; core lifts WB=whole body clean variations, snatch variations, deadlift variations, selected posterior chain exercises; core lifts UB=bench press variations, overhead pressing variations, vertical or horizontal pulling variations; HIR=high-intensity running (>5 m/s); VHIR=very high-intensity running (>6.7 m/s); contact conditioning=rugby-specific contact skills (e.g., wrestling variations); SSC=stretch shortening cycle.

and technical proficiency in a subsequent afternoon session, while additionally lowering force production, capacity, and rate of force application during a potential afternoon strength and power session (refer to chapter 6). The latter aspect may occur due to the potential impairment of the learning process alongside a lessening of the training stimuli needed for maximal neuromuscular adaptation (2, 13).

During back-to-back tournaments, which often have a 6 d turnaround time where match schedules

dictate the training programming, there may be minimal time for HIIT. In contrast, we generally program 1 or 2 HIIT sequences during weeks where a full training week is permitted between tournaments (see table 29.5).

Program Structure and Progression

In Rugby 7s, as with most team sports, there is little periodization that occurs during tournament weeks.

Thursday	Friday	Saturday	Sunday
Rest	Speed qualities	Rugby: low speed clarity	Rest
	Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets×60 s to 2 min, 1 min rest depending on volume and periodization; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Strength/cross training: individual specific needs	
	Conditioning: RST type 4 or 5; 3-6×30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodization of VHIR (team and individual); mostly straight line, hamstring load		
	(1) Type 5: 3-6×all-out 30 s with 2-4 min recovery (consider mostly when no afternoon strength session), possibly interspersed within rugby session; (2) HIIT short interval: type 1; 10 s/10 s, 15 s/15 s, straight line or selected periodized COD/shuttle depending on strength focus for p.m. session (avoid mechanical load); mostly straight line as neuromuscular already overloaded in high-intensity game simulated segments during training.		
	Rugby: craft session or low intensity/low running-volume skills	Rest	
	Strength: WB or LB power or complexes/plyometrics; fast SSC activity (late preseason and lower volume)		

In season, we aim to keep the weekly training contents consistent in terms of a structured taper into tournament weeks (table 29.6). During tournament weeks, we aim to keep the weekly training load nearly constant, with a slightly lower volume during week 2 throughout the World Series and focus on successive tournament preparation. Simply put, the cycle runs as tournament, recovery, preparation, taper, tournament, and so forth. There is a small emphasis on physical development with increases in training volume and intensity during preparation periods between back-to-back tournaments with careful

consideration given to individual player and group monitoring (fitness and fatigue monitoring; e.g., wellness, groin squeeze, neuromuscular, HR) as well as the monitoring of acute and chronic workloads (i.e., acute/chronic ratio (14)).

The format of within- and between-tournament programs (see table 29.5) and program objectives remain similar across the season, with the exception of preparation for stand-alone tournaments such as the Olympic Games or Rugby Sevens World Cup, with a longer and more structured periodization and associated taper.

Table 29.3 $\dot{V}O_2$ and Strength/Speed Development Cycle
(Specific Phase 2, Preparation Phase Between Tournaments;
Typically 1-2 Weeks Duration)

Microcycle		Monday	Tuesday	Wednesday
$\dot{V}O_2$ + strength/ speed develop- ment (specific)++	a.m.	Speed qualities	Strength: UB core lifts	Speed qualities
		Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets $\times$ 60 s to 2 min, 1 min rest depending on volume and peri- odization; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Contact conditioning	Rugby (contact): low-volume running load
		Conditioning: RST type 4 or 5; 3-6 $\times$ 30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodiza- tion of VHIR (team and individual); mostly straight line, hamstring load		Conditioning: repeat effort off-ground conditioning with repeat acceleration over 10, 20, or 30 m (type 4); consider lactate**
		HIR short interval: straight line (type 1) or COD (type 2) 10 s/10 s or 15 s/15 s or shuttle depending on week and periodization		
	p.m.	Rugby: craft session or low intensity/ low running-volume skills	Rest	Speed qualities (acceleration): careful consideration to volume and type of type 4 repeat effort/ acceleration in the morning session and time between morning field session and afternoon session
		Strength: LB core lifts		Strength: LB complexes with acceleration sprints (10-20 m indoor, 3-6 repetitions)

Note that two days are allowed after the lactic session Friday and that the long-bout HIIT is programmed before the day off. Type refers to the physiological target.

COD: change of direction; core lifts LB=squat variations, trap bar variations; core lifts WB=whole body clean variations, snatch variations, deadlift variations, selected posterior chain exercises; core lifts UB=bench press variations, overhead pressing variations, vertical or horizontal pulling variations; HIR=high-intensity running (>5 m/s); VHIR=very high-intensity running (>6.7 m/s); contact conditioning=rugby-specific contact skills (e.g., wrestling variations); SSC=stretch shortening cycle.

Incorporation of HIIT

In the off-season, we often prescribe a heavy loading run-based HIIT program, with the aim of enhancing aerobic capacity and aerobic power prior to the commencement of the preseason. During the preseason, we incorporate HIIT sequences into training sessions either in between technical and tactical sets or repetitions (mostly SSGs or MNRs) or at the end of the session with group or individual modifications made based on fitness and fatigue monitoring or individual load management. The two most important considerations made with respect to selecting the desired types of metabolic and neuromuscular responses to the HIIT sequences during the training week within our program are

1. the demands of the other training components in which HIIT is incorporated, and
2. the individual locomotor loading patterns over the weekly cycle so HIIT can be used to complement or compensate the load arising from training while lessening as much fatigue accumulation as possible to promote subsequent session quality (e.g., afternoon or following day).

It should also be noted that while more optimal within-day programming may exist, our program works within the constraints of coach priorities as well as logistical challenges (e.g., access to fields during visits to foreign countries; see the case study at the end of the chapter), not to mention team and train-

Thursday	Friday	Saturday	Sunday
Rest	Speed qualities	Rugby: low speed clarity	Rest
	Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets×60 s to min, 1 min rest depending on volume and periodization; gradual increase in COD (monitor) throughout preseason and transition between tournaments	Strength/cross training: individual specific needs	
	Conditioning: RST type 4 or 5; 3-6×30-40 m with 12-25 s recovery (1-2 sets throughout session) based on loading and careful periodization of VHIR (team and individual); mostly straight line, hamstring load		
	Usually a half day before departing for travel, no afternoon training sessions. Key posterior strength exercises or injury prevention exercises may be undertaken at completion of the main field session.		
	Rugby: craft session or low-intensity/low running-volume skills	Rest	
	Strength: WB or LB power or complexes/plyometrics; fast SSC activity (late preseason and lower volume)		

ing culture. Therefore, the type of additional HIIT sequences within or at the completion of a rugby training session is given careful consideration, alongside awareness that HIIT sequences can be included in the morning prior to an afternoon strength and power session, often twice a week during the preseason, or in preparation weeks between tournaments. In addition, more challenging HIIT sequences are typically prescribed before a day off or before an off-legs training day to allow for more appropriate recovery. Consideration is therefore given to minimize possible interference effects to maximize adaptation (see chapter 6).

In season or during preparation weeks between tournaments, the amount of weekly HIIT prescribed takes into account individual overall playing volume and loading history over the previous few days and weeks. In Rugby 7s, our priority is to ensure all players are sufficiently loaded so as to

maintain fitness and be prepared for the required intensity during match play across a tournament, and in particular the high-intensity running and acceleration, deceleration, and change of direction volumes (mechanical work), as well as tapering enough to feel fresh entering a tournament. This requires a highly individualized HIIT prescription that can be achieved using a combination of different HIIT types and formats.

Sample Training Programs

Tables 29.1, 29.2, 29.3, and 29.5 provide examples of HIIT programming during different phases of training, i.e., preseason (tables 29.1 to 29.3) and in season (table 29.5). Note the distribution of the different HIIT types and loading in relation to the microcycle's objectives (table 29.6).

Table 29.4 Example Small-Sided Games (SSGs) or Manipulated Numbers and Rules (MNRs)

Drill	Field size (m)	Number of players	Duration of set (min)	m/min	HIR	VHIR	VHIR%	Average HR (% of max)	Max HR (% of max)	Accelerations/ min (>2.5 m/s)	Decelerations/ min (<2.5 m/s)
Contested plug defense drill	55×55	6v3	1	191.1	48	11	5.0	88.6	94.5	3.7	2.8
		6v4	1.5	186.7	74	17	5.6	90.0	94.8	2.4	1.7
SSG offside touch	60×60	7v4	1.5	204.2	98	26	7.6	91	96	2.1	1.6
	100×70	6v6	3	196.5	205	29	5.3	175	95.5	0.4	0.4
		7v7	2	182.5	97	19	4.9	174	94.4	0.8	0.2
			2.5	188.6	111	12	2.8	175	92.4	0.4	0.3
			3	187.8	196	25	4.2	169	92.8	0.3	0.3
			3.5	167.9	210	36	6.4	171	92.6	0.8	0.5
	50×70	3v3	2.5	154.9	53	7	2.0	168	93.2	1.4	0.9
			3	157.5	68	1	1.0	164	90.6	2.7	1.4
		4v4	4	155.4	68	3	0.5	N/A	N/A	1.4	0.7
	60×38	4v4	2	185.7	100	15	4.2	174	93.8	0.8	0.4
70×30			2.5	174.9	105	15	3.3	169	94.2	0.8	0.6
		4v4	2	176.9	137	41	11.4	160	89.8	0.9	1.3
			2.5	150.0	91	30	8.4	168	91.1	0.7	0.8
	70×50	5v5	2	176.5	130	20	5.1	182	96.4	0.8	0.5
			2.5	177.8	102	12	2.8	176	96.3	0.9	0.6

HIR: high-intensity running >5 m/s; VHIR: very high-intensity running >6.7 m/s; VHIR%: % very high-intensity running >6.7 m/s in relation to volume; average HR: average heart rate; N/A: not available

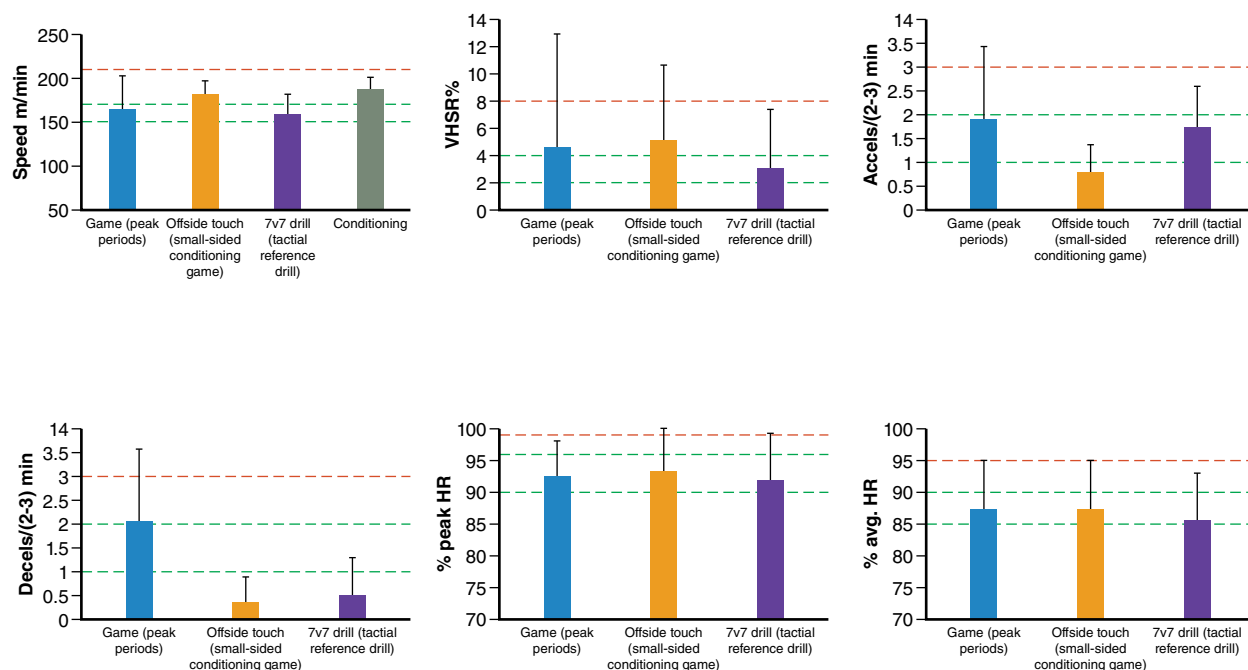


Figure 29.7 Comparison of physiological and performance demands between a typical game, a SSG (offside touch), a tactical drill (7v7), and HIIT conditioning sessions. Offside touch = 2 min duration; 7v7 drill = 2 min duration; game (peak periods) and worst case scenarios calculated from comparison of GPS and time-stamped video files; conditioning = average m/min for conditioning drills undertaken (e.g., 15 s on/15 s off, 20 s/20 s, 30 s/30 s, 45 s/15 s); red line indicates highest values seen with regular frequency in match play. Green lines indicate range of most peak activity occurring. Analysis determined through visual inspection of histograms. VHSR% = percentage high-speed running (>6.7 m/s) in relation to total volume. Accels/min = accelerations per minute (>2.5 m/s). Decels/min = decelerations per minute (<-2.5 m/s). Peak and avg HR = peak and average heart rate reached during the different drills.

Table 29.5 Maintenance (In Season; Tournament Week 1, Typically Follows Long-Haul Travel)

Microcycle (physical emphasis)	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Competitive (in-season tournament week 1)	Travel, arrive	Flush session	Speed qualities	Rest	Game preparation	Tournament	Tournament
	Recovery or flush session	Submaximal HRR test: 3-4 × 3-4 min submaximal run intervals at increasing intensity; 3-4 strides over 40-50 m (maximum 70% maxi- mum velocity), light skills or game with low running volume, HIR, and VHIR	Rugby: includes type 1 or 2; 7v7, 7v6; 4-8 sets × 60 s to 2 min; 1 min rest depending on volume or periodization; gradual increase in COD (monitor) through- out preseason and transition between tournaments		Rugby: low- volume rugby skills with approximately 10 min of lower-intensity tactical sequences		
	Recovery + mobility, trigger, or roll session throughout the day depending on travel schedule; continue adjustment to time zone changes and associated strategies to minimize jet lag especially after long-haul travel	Strength: UB core lifts		Strength: UB × 30 min priming session	Rest		
		Rest	Strength: WB or LB power or complexes or plyometrics; fast SSC activity; some exposure to core LB exercises in week 1				

Note that two days are allowed after the lactic session Friday and that the long-bout HIIT is programmed before the day off. Type refers to the physiological target.

COD: change of direction; core lifts LB = squat variations, trap bar variations; core lifts WB = whole body clean variations, snatch variations, deadlift variations, selected posterior chain exercises; core lifts UB = bench press variations, overhead pressing variations, vertical or horizontal pulling variations; HIR = high-intensity running (>5 m/s); VHIR = very high-intensity running (>6.7 m/s); contact conditioning = rugby-specific contact skills (e.g., wrestling variations); SSC = stretch shortening cycle; HRR = submaximal heart rate response test that involves 4 min continuous running at 12 km/h with 50 m intervals undertaken every 15 s.

Table 29.6 Example of Typical Periodization Between Tournaments

Week	Periodized load		Total distance (m)	HIR (m)	VHIR (m)
Week 3	Medium to high volume	Mean±SD	19216±1719	3097±1072	512±48
		Range	17563-21345	1663-4491	451-581
Week 2	High volume	Mean±SD	15581±2134	2461±581	746±153
		Range	13681-18714	1594-3090	553-917
Week 1	Medium volume	Mean±SD	13015±576	1788±243	463±59
		Range	12525-14015	1566-2193	386-526
T1	Tournament	Mean±SD	17505±1912	2278±496	578±145
		Range	15139-19536	1588-2802	374-752
T2	Tournament	Mean±SD	15840±3462	1911±583	477±181
		Range	11031-19791	1136-2469	244-671

HIR=>5 m/s; VHIR=>6.7 m/s

CASE STUDY

The following story describes a series of events that led to a minor hamstring injury and subsequent scenarios leading to the return of a key player to competition only 2 wk later. To provide some background for this story, we begin mid-season following our Las Vegas World Series tournament, where this particular player had been given an individual session to undertake between this tournament and Vancouver (March 12–14, 2016; see story timeline in figure 29.8). We had not been able to obtain GPS data from the Vancouver tournament due to its indoor nature, nevertheless the player was required to undertake an easy progressive running session that included some strides over 50 m to a self-regulated intensity of approximately 80%. This typical flush session involves about 3 to 4 km in total running volume, with about 30 to 50 m of high-intensity running (>6.7 m/s) gained while undertaking a series of strides over 50 m. On this occasion, the player did not complete the individual flush session for reasons unknown to us, further highlighting the challenges of even professional athletes taking responsibility for the undertaking of individual external training themselves.

The following week, upon commencement of our main preparation week for the Hong Kong Sevens, the player produced far more high-intensity running during selected drills in the first session than was planned or as drill predictions had indicated. This meant a significant spike in high-intensity run load following the previous tournament week in Vancouver (which had been completed 8 d earlier). The player also missed the following few days of training with illness. Upon returning 4 d later, the player was required to undertake a supervised extended flush session, including a type 2 targeted short-interval HIIT session involving 10 min of 30 s running at 95% VIFT with 30 s passive recovery over an 80 m shuttle, in order to progress run volume prior to returning to the main squad and performing in a high-intensity rugby session.

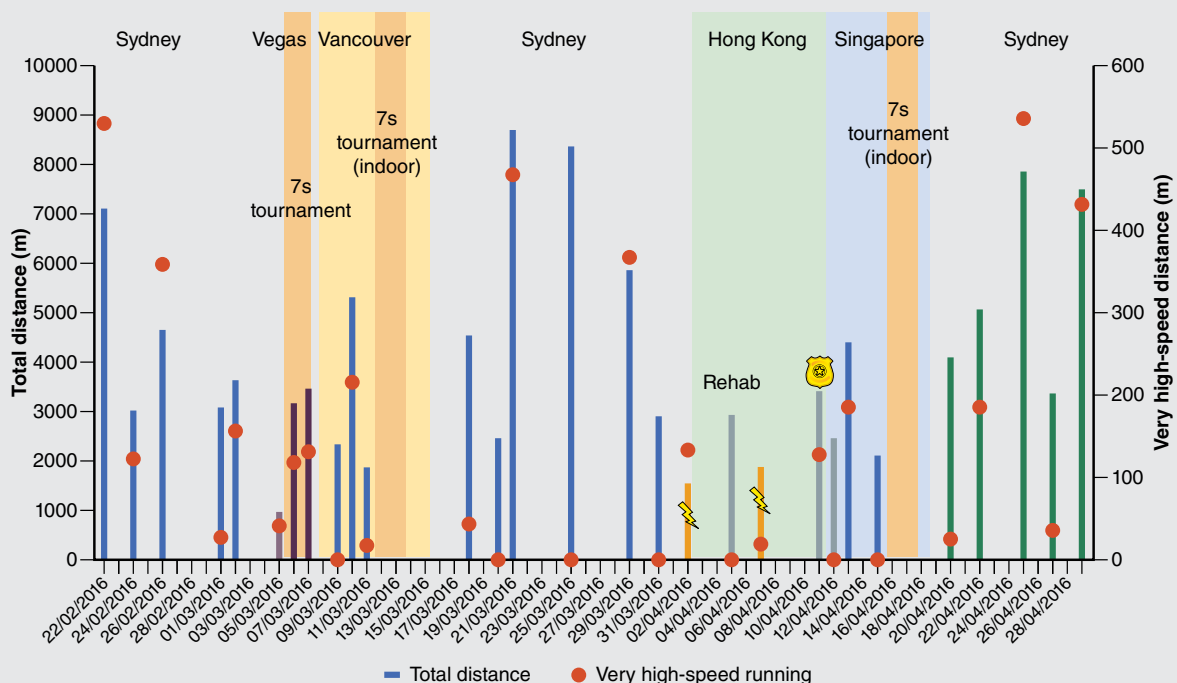


Figure 29.8 Total and very high-speed distance (>6.7 m/s) shown. No GPS data available for both indoor tournaments. Session of injury and reinjury occurrence (slipping during rehab) are shown in yellow. Rehab sessions in green. The session in Hong Kong seeking access to fields to undertake progression of volume and exposure to speed or high-intensity running is shown by the badge.

In the final preparation week prior to departure for Hong Kong, the player completed the normal main Rugby 7s training session earlier in the week, typical of a session in the week prior to a tournament (5-6 km total distance; 350-400 m of high-intensity running [>6.7 m/s]). In our final main training session prior to departure for the Hong Kong Sevens tournament (a marquee tournament on the World Series Sevens circuit), the player experienced a minor hamstring aggravation during his final set of high-speed flys (20 m roll to 30 m sprint).

Following clinical testing and assessment, it was decided that although the strain was minor, he would still travel and complete his rehabilitation and return-to-play protocol alongside myself and our head physiotherapist while we were in Hong Kong, in order to give him the best opportunity to play in the following tournament in Singapore (6 d after the completion of the Hong Kong Sevens tournament). As a performance and medical team, we acknowledged that even with the diagnosis of a minor hamstring strain, time wasn't on our side to get our player ready to perform in the Singapore tournament, but collectively decided that his best chance would be to travel with the team. With a carefully planned progression of volume, high-intensity running, and gradual increases in mechanical load over the 2 wk, we felt the player could potentially be ready to play in the Singapore tournament, even if it meant having him available for day 2 of the tournament with controlled game time on day 1 if required.

The day following arrival in Hong Kong, the player completed a normal flush session as described above, which also served as a controlled assessment of hamstring function and pain. Two days later, a planned progression of volume and high-intensity running was interrupted when the player slipped and felt a small degree of pain at the hamstring injury site during a short footwork warm-up progression drill (low mechanical load). As one can imagine, we were extremely concerned. However, the player was reassessed 24 hours later, presented well clinically, and following recalibration of our high-speed running, changes of direction, and mechanical load progressions, the player remained on track and abreast of all technical and tactical requirements for the upcoming tournament. On the Monday prior to the Singapore tournament, with the remainder of the squad undertaking a recovery session in Hong Kong prior to our Singapore flight, the injured player and I were forced to search for a field or open space in Hong Kong to ensure our targeted progression of volume and high-intensity running stayed on track. This was a crucial session to prepare for the main training session for the week on the Wednesday in Singapore (2 d later) prior to the Saturday tournament. We caught a taxi from the hotel to a proposed available and prebooked field that was supposed to be available for us, but when we arrived, we were unable to find the pitch. After unsuccessfully searching for another 20 min, we jumped a fence into a closed private sports club, commenced a basic warm-up and agility progression and into our prescribed HIIT program, set up at the far end of the field away from the clubhouse and entrance in case we were disturbed. As expected, a staff member at the sports club approached me, at which time I told the player to continue the HIIT set (another 4 min) while I pleaded ignorance to the staff member and kept him occupied. While we were kindly asked to leave the premises, it was not until the player completed a good portion of his prescribed HIIT set. However, with some volume, reactive agility, footwork, strides, and further progression of VHIR still required, we embarked on another venture to source field space to complete the remainder of the planned session. Fortunately, we were able to find another enclosed sports stadium, and given our time restraints, made the decision to jump yet another fence to complete the remainder of the training session. Our player immediately commenced the remainder of the session himself, knowing that I would continue to provide interference and distraction for incoming security guards and groundskeepers. Following a second encounter, the session was completed and we quickly hailed a taxi back to the hotel before flying out to Singapore. The player was subsequently able to complete the main training session of the

(continued)

CASE STUDY *(CONTINUED)*

week 2 days later with the team and performed well during the Singapore Sevens World Series tournament.

I was extremely relieved that the player could compete at the tournament after I and our performance and medical team encouraged the travel decision. However, I was disappointed in myself for not minimizing player risk during his rehabilitation progressions. I had set the warm-up and reactive agility station in an area of the ground that presented as a wet and uneven surface, and missed seeing the player perform this part of the session in runners and not boots. The experience not only reminded me of the importance of paying attention to detail required when undertaking a crucial rehabilitation session with time restraints, but also the importance of monitoring progressions of training load and VHIR at any time in the program. Each day and each session becomes more and more critical as a tournament or major competition looms.